HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented complex circuit analysis problems for AI model evaluation, focusing on analog and mixed-signal design challenges.
- Critiqued AI-generated schematics and component selections against professional engineering standards, identifying and flagging invalid topologies and unphysical results.
- Compared performance between two frontier AI model versions on identical electronic engineering prompts, providing detailed reasoning for superiority and specific shortcomings.
- Developed realistic engineering problems from personal practice in circuit specification and control loop stabilization for AI model training and validation.
- Flagged concrete AI failures with evidence, including wrong assumptions, sign/unit errors, and fabricated datasheet values, ensuring technical accuracy.
- Contributed expertise across signal processing and power electronics domains to refine AI model outputs for specialized engineering applications.
- Containerized Python-based AI evaluation tooling and deployed to a Kubernetes cluster (EKS) for scalable and repeatable model assessment across diverse engineering prompts.

SKILLS

Core Engineering Concepts – Circuit Analysis, Control Systems, Power Electronics, Signal Processing, Analog Circuits, RF Circuits, Mixed-Signal Circuits, Embedded Systems, Board-Level Design, Photonics, Devices and Materials, Electromagnetics, Antennas

AI & ML Tools – AI Model Evaluation, Frontier AI Models, Prompt Engineering, Model Comparison, Technical Critique, Data Annotation

Software & Scripting – Python, Shell Scripting, Linux

Cloud & DevOps – Docker, Kubernetes, AWS (EC2, S3, Lambda, CloudWatch), Jenkins

Data & Analytics – SQL, Power BI, Tableau, R / RStudio, pandas, NumPy, scikit-learn

Engineering Design & Simulation – Circuit Design, Simulation, Debugging, Schematics, Component Selection, Calculations

Professional Skills – English Fluency, Technical Writing, Problem Solving, Cross-functional Collaboration

PROJECTS

AI Model Evaluation for Analog Circuit Design

- Developed a suite of challenging analog circuit design problems, including component sizing and stability analysis, to test AI model capabilities.
- Executed Python scripts to interface with frontier AI models, feeding them engineering prompts and systematically evaluating the generated schematics and calculations.
- Authored detailed critiques of AI outputs, specifically identifying sign errors, unphysical component values, and incorrect circuit topologies, contributing to model refinement.

Control Systems AI Trainer (HCL Technologies)

- Designed and implemented realistic control loop stabilization problems, drawing from enterprise systems support experience, for an AI training initiative.
- Compared output from two distinct AI model versions against a control systems expert's benchmark, documenting specific failures and suggesting areas for improvement.
- Provided feedback on AI-generated derivations and adherence to system-level design tradeoffs, ensuring technical accuracy in the training data.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)